

EmbeddedLend: API-First SME Underwriting Layer

The system contract for the unbuilt bridge — consent-based mobile-money underwriting on top of BNR Open Finance (data rail) and eKash (money rail). API-only scope; UI deferred.

EmbeddedLend (rename at PRD stage)

Rwanda → East African Community

PRD (architecture) → MVP slice flagged → Demo build

v1.0 · August 2026

Problem Statement, Deep Research, Qianhe Autopsy

Abstract

EXECUTIVE SUMMARY

EmbeddedLend is an API-first SME underwriting service for the "missing middle" of Rwanda's credit market. It converts consented mobile-money cash-flow into defensible, explainable loan decisions, sitting on top of the two national rails: BNR Open Finance (the data rail, voluntary phase now) and eKash (the money rail, live July 2026).

This PRD specifies the full architecture — a layered stack from consent and data access, through deterministic signal extraction and scoring, to loan operations and eKash money movement, with compliance and audit as a vertical slice. The MVP slice (flagged throughout) is the "one livable room" that proves the core bet: one consent flow, one data source, the scoring engine, the eKash connector, and the core endpoints. UI productization is explicitly out of scope; a demo interface is built separately as an outreach artifact.

Contents

1. Goals & Non-Goals
2. Personas (Primary)
3. Architecture Overview
4. Data Flow (Borrower Journey)
5. Component Map
6. API Surface Sketch
7. Compliance Architecture
8. MVP Slice (Flagged)
9. Tech Stack
10. Evidence Ledger (PRD-specific)

1. Goals & Non-Goals

Goals

- Deliver an **API-first SME underwriting service** enabling licensed lenders to originate, underwrite, and service business loans for the "missing middle" — MSMEs too informal for collateral-based bank lending and too business-like for consumer micro-loans.
- Turn consented **mobile-money cash-flow** into a **defensible, explainable loan decision** — the unbuilt bridge between the data rail (BNR Open Finance) and the money rail (eKash).
- Make the regulator comfortable with **more lending, not less**: consent-first data pulls, audit trails, explainable decisions, data residency.

Non-Goals (explicitly out of scope)

- **No UI/UX productization** in this PRD — no lender portal screens, no onboarding flows. UI is deferred pending primary research on the player landscape; a demo interface is built separately as an outreach artifact.
- No direct consumer lending, no deposits, no holding a loan book (the Qianhe lesson — we never take credit risk; the lender remains the regulated entity).
- No ML model training or black-box scoring in the MVP — deterministic, transparent scoring first.
- No multi-bureau, multi-rail integration in the MVP.

2. Personas (Primary — who pays)

Persona	Description	Consumes
NDFSP (non-deposit-taking financial service provider)	One of 80+ licensed digital lenders in Rwanda (e.g. Zamuka, IWACU, EXUUS). Tech-poor, small, needs plug-and-play origination + scoring.	Full stack: consent, data pull, score, decision, loan, disbursement, repayment
MFI / SACCO	Extends credit to women and micro-enterprises; runs on field-officer apps and paper.	Scoring API + decision API (brings own distribution)
Vertical SaaS platform	E-commerce, logistics, agritech, retail POS serving merchants (e.g. Kayko corridor). Wants embedded working capital without becoming a lender.	Embedded credit flow: score + decision + loan + eKash disbursement inside their product

Secondary (future): regional banks preferring white-label APIs; digital lenders expanding into Rwanda.

3. Architecture Overview

A layered, API-first microservice integration layer. The lender's core banking system stays the system of record; our services sit beside it.

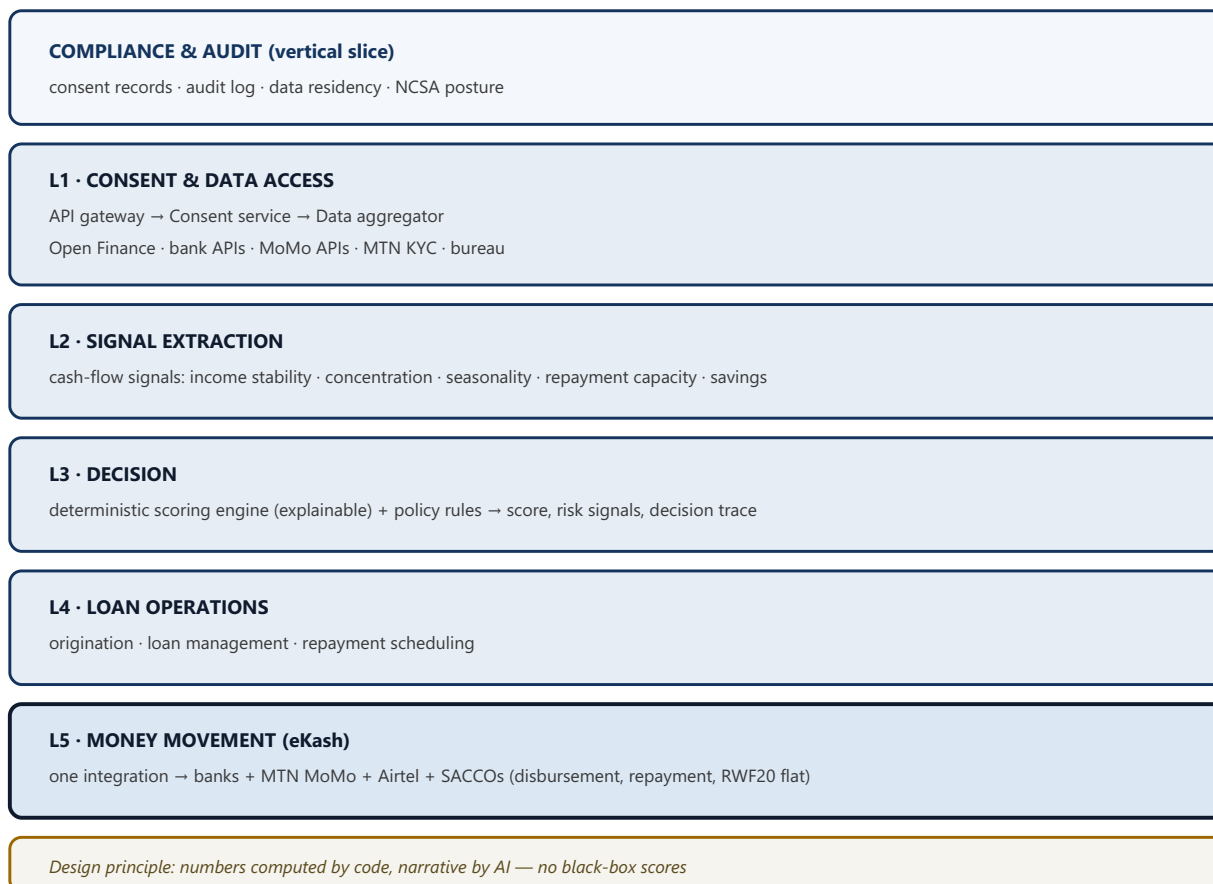


Figure 1 — EmbeddedLend layered architecture (MVP slice shaded in Figure 3)

4. Data Flow (Borrower Journey)

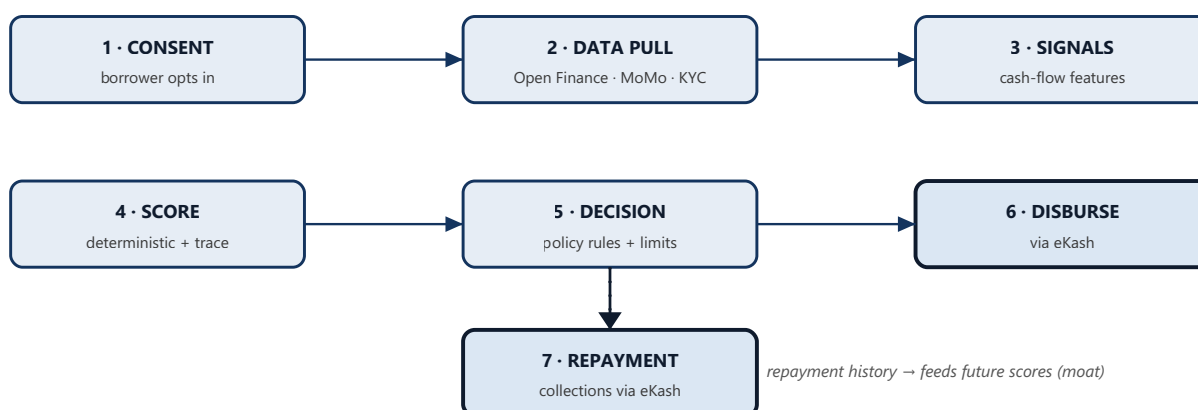


Figure 2 — Data flow: consent → data → signals → score → decision → eKash disbursement → repayment

5. Component Map

Component	Responsibility
API Gateway	AuthN/AuthZ, rate limiting, tenant isolation per lender
Consent Service	Capture, record, revoke consent; audit-gated
Data Aggregator	Normalize Open Finance / bank / MoMo / KYC / bureau data into one schema
Signal Extractor	Compute cash-flow signals (deterministic)
Scoring Engine	Deterministic, explainable score + risk signals
Decision Service	Policy rules, limits, decision trace
Loan Manager	Origination, loan lifecycle, repayment scheduling
eKash Connector	Disbursement + repayment over the national rail
Audit Store	Tamper-proof decision + consent audit trail

6. API Surface Sketch

Endpoint	Purpose
POST /v1/consent	Create borrower data-sharing consent
GET /v1/data/summary	Fetch consented, aggregated data + cash-flow signals
POST /v1/score	Compute deterministic score + risk signals
POST /v1/decisions	Apply policy rules → decision + trace
POST /v1/loans	Originate a loan
POST /v1/disbursements	Disburse via eKash
POST /v1/repayments	Record/collect repayment via eKash

7. Compliance Architecture

- **Consent-first** data pulls with purpose limitation (Law 058/2021 + GDPR-style rights: explainability, correction, portability).
- **NCSA posture** — data processor/controller certification path per the Open Finance roadmap.
- **Data residency** in jurisdiction; tamper-proof decision traces; full audit log.
- **Explainability** — every decision carries its risk signals and reasoning; the "numbers computed, narrative by AI" principle keeps the LLM as narrator, never the decider.

8. MVP Slice (Flagged — what we build first)

MVP — THE "ONE LIVABLE ROOM"

The minimum cut that proves the core bet: a lender can plug in, get a borrower's mobile-money data, get a defensible score, and disburse/repay via eKash.

In MVP	Out of MVP (later)
Consent flow (one)	Loan-management dashboards
One data source (mobile-money cash-flow)	Multiple bureaus
Scoring engine (deterministic, explainable)	KYC vendor integrations
Decision service + trace	ML model training
eKash connector (disburse + repay)	Fancy reporting / analytics
Core endpoints: consent, data/summary, score, decisions, loans, disbursements, repayments	Multi-country expansion

MVP success metric: a pilot lender originates a real business loan to a thin-file borrower using consented mobile-money data, with a defensible decision trace — proving willingness-to-pay and data-layer feasibility.

MVP coverage across the stack

L1 Consent & Data Access (one source)

L2 Signal Extraction (cash-flow signals)

L3 Decision (scoring + policy + trace)

L4 Loan Operations (minimal: origination + lifecycle)

L5 Money Movement (eKash connector)

Compliance & Audit (minimum: consent + trace + residency)

Shaded = shipped in MVP. Plain = deferred.

Figure 3 — MVP slice flagged across the architecture

9. Tech Stack

- **Backend:** Node/TypeScript, Postgres
- **Infra:** API-first microservices, containerized
- **Money rail:** eKash (one integration → whole ecosystem)
- **Data rail:** Open Finance / bank / MoMo consent APIs; MTN KYC API

- **Decision:** deterministic scoring engine (pure compute) + optional LLM narration layer (config at deploy time)
- **Budget/timeline:** side-project scale (~\$200–800/mo); MVP ~3–4 months part-time; pilot with 1–2 lenders is the real milestone

10. Evidence Ledger (PRD-specific)

Claim	Source	Grade
eKash = one integration for the whole ecosystem (banks, MTN MoMo, Airtel, SACCOs, MFIs), RWF20 flat, RWF10m cap	New Times, African Business, RISA (Jul 2026)	✓ Observed
eKash is a payments rail only — no credit scoring/loan decisioning	Launch coverage	✓ Observed
BNR Open Finance objective #1 = alternative scoring for MSMEs; voluntary Phase 1 now, mandatory Phase 2 (2026-28), central platform Phase 4 w/ private partner	BNR Open Finance Position Paper	✓ Observed
MTN KYC API exists, consent-gated, use case "digital lending"	MTN Rwanda	✓ Observed
BK Open API (Mar 2026) enables payments/credit access/data exchange	KT Press	✓ Observed
80+ licensed NDFSPs; Jali Finance: "credit reference systems remain a structural challenge"	BNR list; allAfrica (Apr 2026)	✓ Observed
Deterministic-compute + LLM-narration pattern validated in finance (FinRobot)	FinRobot repo / AI4Finance	🟡 Secondary
Willingness-to-pay by Rwandan lenders	No interviews yet	? Assumption

Method: architecture derived from the problem statement (problem-embeddedlend.md) and deep research (research-embeddedlend.md); facts graded in Section 10. The MVP slice is the first implementable increment; UI productization and primary research on the player landscape run in parallel. Open question: lender willingness-to-pay, to be validated in the Field stage.